

Advanced Smart Filtration Techniques for Real-time Air Quality Monitoring and Improvement, along with a machine learning-based alert system

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Abstract—Poor air quality due to industrialization and urbanization causes respiratory problems and health concerns. Nearly 6.67 million people had died due to air pollution globally. To reduce we need to bring a new regulations as well as device for air pollutant purification. The proposed system has the stability of purification. Continuous monitoring and filtering are essential. This system integrates sensors and filters to provide real-time air quality data, using machine learning for accuracy. SMS alerts provide early warnings for timely action, increasing personal health awareness and regulatory compliance. Overall, it promotes better settings by reducing the negative effects of air pollution on public health and the environment.

Index Terms—Air quality analysis, Sensors, SMS

I. INTRODUCTION

Air pollution is a significant global concern, posing risks to human health and the environment. We need to reduce the air pollution, but we are trying to implement by setting new rules. This leads to change in the environment after few years. As of now we need to get a breathable air to this we need to purify the polluted air. This research aims to develop an air quality analysis and filtering system using MQ-02 and MQ-135 gas sensors, temperature and humidity sensors, and multiple filters including fine dust, cold catalyst, activated carbon, HEPA, sterilization cotton, and negative ion filters. The system utilizes an Arduino microcontroller, three fans for air circulation, GSM communication for sending SMS alerts, and data collection. Additionally, machine learning algorithms are applied to enhance accuracy in analyzing sensor data. This research contributes to the field of air quality monitoring and control, with potential implications for public health and sustainable living.

A. Background information on air pollution

Air pollution is a pervasive issue that has detrimental impacts on both human well being and the natural environment. To comprehend and mitigate the effects of air pollution it is imperative to engage in thorough monitoring of air quality and employ effective water filtration techniques. Monitoring is aided by integrated gas sensors (MQ-02 and MQ-135), temperature and humidity sensors, and other sensors. Fine dust, cold catalyst, activated carbon, HEPA, sterilizing cotton, and negative ion filters are some examples of filters that offer purification. Weather forecasts are more accurate thanks to machine learning algorithms, and the GSM connection enables remote SMS monitoring. Our project aim is to create a system that safeguards the environment and public health while adhering to air quality analysis and approval requirements.

B. Problem Statement and Objectives

The problem is that air pollution is increasing, and good air monitoring and detection systems are needed. Existing solutions often lack accuracy and real-time monitoring capabilities [1]. This study intends to use MQ-02 and MQ-135 gas sensors, temperature and humidity sensors, dust filters, cooling catalysts, activated carbon, HEPA, sterilization cotton, and negative ions. The system will incorporate GSM to enable remote data delivery via SMS. These research focus on the system's accuracy using machine learning algorithms, and they provide dependable methods for tracking and enhancing indoor air quality.

II. LITERATURE REVIEW

Many research is done based on air quality monitoring by using gas detector sensor such as the MQ-02 and MQ-135 sensor, the sensor collects data which monitors the pollutants level such as nitrogen dioxide and carbon monoxide [2][3]. The air quality will be affect based on the temperature and humidity sensor. It is commonly used in air quality analysis since they offer the data required to comprehend environmental factors. Air filters have been analyzed for their capacity to eliminate particular contaminants and enhance air quality, including fine, water, HEPA, disinfectants, and unfiltered filters. Examine how GSM technology may be used to communicate air quality and send large amounts of data for data gathering and analysis. Air monitoring depends on machine learning algorithms to process sensor data, identify patterns, and measure accuracy to have a better understanding of air pollution and the associated healthy consumption [4].

A. Review of existing studies on sensors

The recent studies have provided data of gas sensors, including the MQ-02 and MQ-135, detect pollutants like nitrogen dioxide and carbon monoxide. When it comes to detecting and tracking oil levels, these sensors have a good accuracy. The ability of humidity and temperature sensors, such as DTH11, to measure both relative humidity and temperature was also carefully analyzed. Many applications, such as managing and monitoring air quality, have given best value of these sensors. In general, the research indicates that MQ-02, MQ-135, and DTH11 sensors are useful for measuring air quality, and it highlights the significance of these sensors for environmental monitoring related to human consumption, health, and well-being.

B. Examination of different types of air filters

- Fine Dust Filter: Intended to trap minute particles like pollen and dust.
- Cold catalyst filter: catalyzes and degrades contaminants like formaldehyde into harmless byproducts.
- Activated Carbon Filter: Uses a porous material to absorb smoke, odors, and chemicals.
- HEPA filter: Effective against dust and animal dander.
- Sterilization cotton: contains bacteria such as silver ions, which can kill bacteria and viruses.

These filters are attached to air purifiers to improve air quality by removing various pollutants and allergens.

C. Review of previous works on integrating GSM

In previous work, GSM integration for data transmission to remote air quality and vision systems showed promise for real-time monitoring and control [5]. The use of GSM technology provides remote access to quality weather data by sending sensor data via SMS. Studies show the advantages of GSM in terms of coverage, reliability and ease of use. In addition, researchers have shown that the GSM module is compatible with Arduino-based systems and provides uninterrupted communication capability. However, more research is needed to

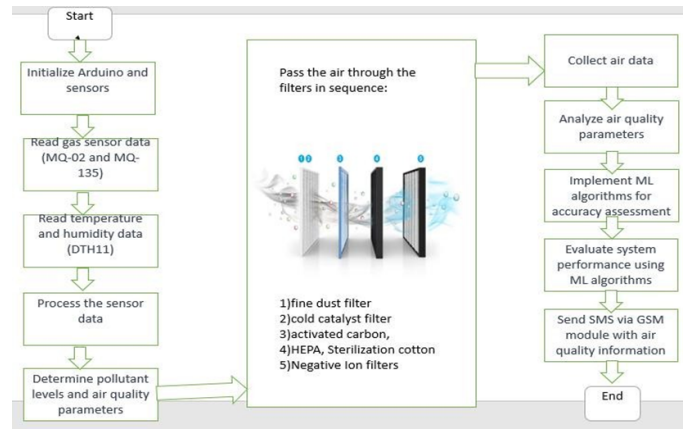


Fig. 1. Flowchart

improve data transmission efficiency and discover the best communication methods for remote monitoring.

III. METHODOLOGY

In our system we have multiple sensors to collect the real time data based on it the training of data is been done in all certain areas of different terrain (Hill station, metropolitan city, rural area). We need to filter the polluted air to get fresh air, to stay healthy. We have explained in detail in below section.

The system starts by initialising the Arduino microcontroller and the sensors that will collect air quality data. Air is then routed through a series of filters in order. These filters are presumably designed to target specific pollutants. The following filters are used: fine dust filter, cold catalyst filter, activated carbon, HEPA filter, sterilisation cotton, and negative ion filters. After the air is filtered, data is collected from a variety of sensors, including gas sensors (MQ-02 and MQ-135), which are likely to detect the presence of specific gases in the air, and a temperature and humidity sensor (DTH11), which measures the temperature and humidity of the air. The data gathered from the sensors is then processed. The processed data is used to assess air quality parameters. This most likely entails converting sensor readings into useful air quality measurements. Machine learning algorithms are then used to determine the accuracy of air quality measurements. The system calculates pollution levels and overall air quality based on processed and analysed sensor data. Machine learning algorithms are used again, but this time to assess the overall performance of the air quality monitoring system. Finally, an SMS with air quality information is sent using a GSM module.

The explained process flow is depicted in the Fig. 1.

A. Arduino

The project uses an Arduino board as the main microcontroller for data acquisition and control. The gas sensor (MQ-02 and MQ-135) and temperature and humidity sensor (DTH11) are connected to the Arduino board.

B. Gas Sensor and Temperature Humidity Sensor

Connect the MQ-02 gas sensor and MQ-135 gas sensor to the digital or analog pins of the Arduino board. Connect the DTH11 Temperature Humidity Sensor to a digital or analog pin on the Arduino board.

C. Filter screen

A. The air filtration system includes various components such as a dust filter screen. Cooling catalyst filter screen. Activated carbon filter screen, HEPA filter screen, sterilization cotton, and negative ion filter screen integrate filters into the airflow to capture and remove certain contaminants. This ensures that all filters are properly connected and placed.

D. Fan

Connect three fans to the Arduino board to create airflow for process air. The uses a relay module to control fan operation. The connects the fan to a suitable power source such as a 9V battery.

E. Battery

A. The uses a 9V battery to power the fan. Depending on the power requirement, connect the batteries to the fans in a parallel or series configuration.

F. GSM Module

A. Connect the GSM module to the Arduino board to send data remotely via SMS. The powers the GSM module with two 9V batteries or a suitable power supply. Establish correct communication between Arduino and the GSM module.

G. Write Data

Data write technique in Arduino code to read measured values at regular intervals. The captures sensor readings from fuel sensors and temperature and humidity sensors. Store archived data in a different or appropriate file for further analysis.

H. GSM Integration

A. Code has been added to Arduino Sketch to send sensor data via SMS via the GSM module. Configures the GSM module to establish a mobile connection and send an SMS with text.

I. Machine Learning Algorithms

The utilization of machine learning algorithms for data analysis and accuracy assessment can yield substantial benefits. By employing data collection as input to train and test these algorithms we are able to effectively evaluate their performance and accuracy in predicting air quality metrics or identifying patterns and anomalies. This approach allows us to make informed decisions based on reliable information. Ensuring a high level of precision in our analyses.

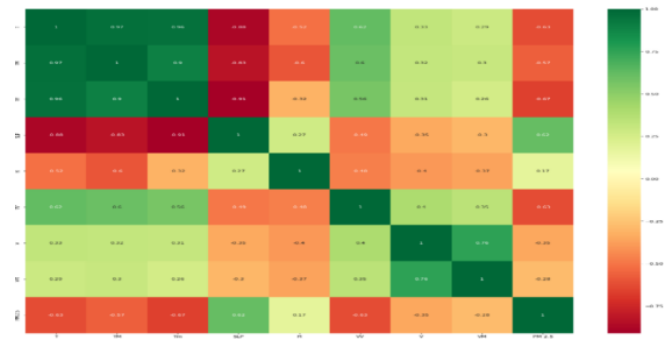


Fig. 2. Coorelation of Heatmap

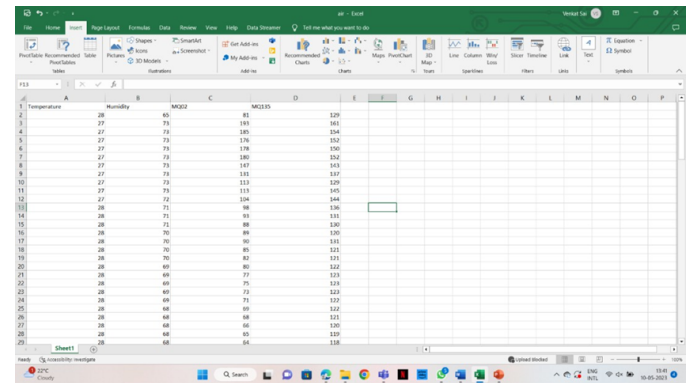


Fig. 3. Real-time Data

J. Corelation of Heatmap

It selects the most relevant features from the database using `cormat.index`, which can calculate the correlation corresponding to the features in the dataframe `df`. It then sets the size of the image to 20 x 20 and uses the `sns.heatmap()` function to draw the heatmap. The `annot=True` parameter specifies that the correlation coefficient should be included in the heatmap. The parameter `Cmap='RdYlGn'` sets the mapping to be used for the heatmap.

Fig. 2 represents the Correlation Heatmap. Fig. 3 illustrates the real time data collected from the sensors.

The bar graph in Fig. 4 depicts the data gathered from our sensors. Fig. 5 illustrates the line plot that portrays the data collected from the sensors.

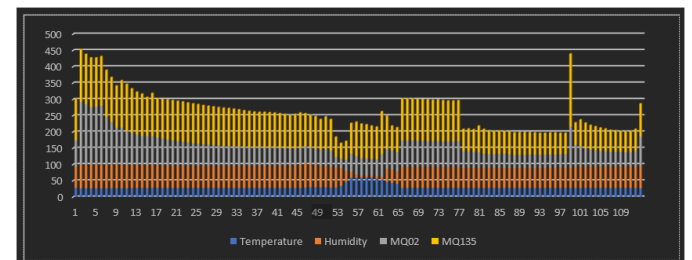


Fig. 4. Bar Graph

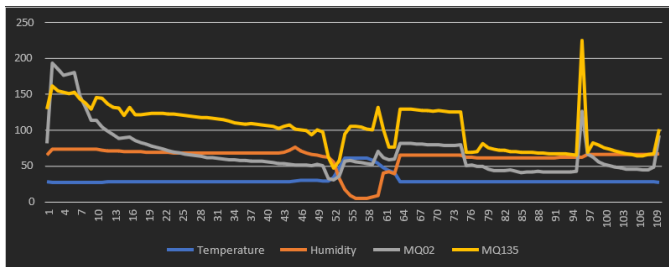


Fig. 5. Line plot

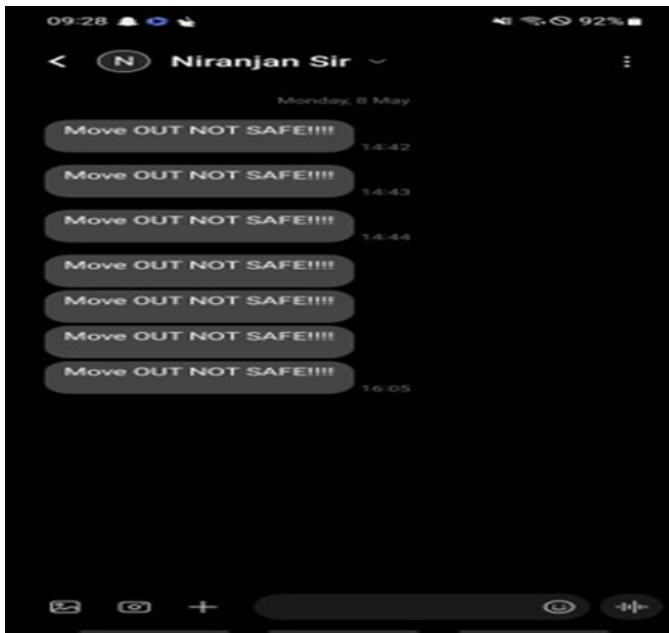


Fig. 6. Message alerts

Fig. 6 depicts the message alerts received on the android device.

IV. RESULTS AND DISCUSSION

A. Presentation and analysis of air quality data collected from gas sensors and temperature humidity sensors:

Describe the data collected from the gas sensors (MQ-02, MQ-135) and temperature humidity sensor (DTH11). Present the trends, patterns, and variations observed in the air quality data. Analyze the relationship between different pollutants and their concentrations. Discuss any significant findings or correlations identified in the data. The data collected from the sensors are been stored in the data base for training the data and prediction through the ML algorithm. Based on the process we are able to analyze the pollutants in the air.

B. Evaluation of the effectiveness of each filter in removing specific pollutants:

Assesses the performance of each filter (fine dust, cold catalyst, activated carbon, HEPA, sterilization cotton, negative ion) in removing targeted pollutants. Compares the reduction in pollutant levels before and after passing through each filter.

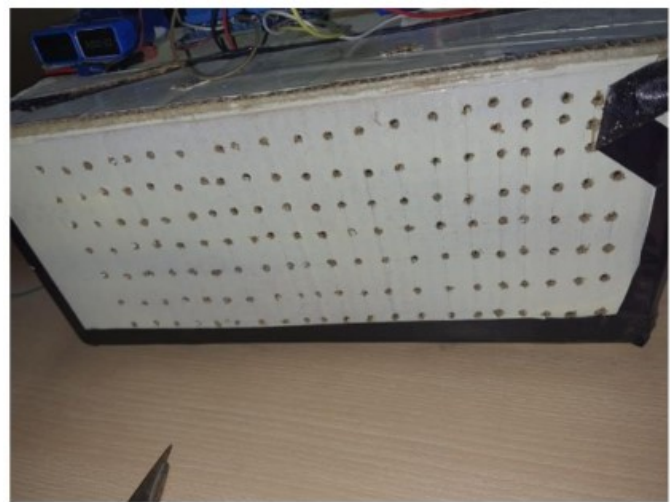


Fig. 7. Front View of Proposed System

Discusses how each filter improves air quality. Highlights any problems or restrictions identified during the filtering process. Based on the previous stage the filtration will be done and produce fresh air, this process of filtration is done in 7 stages of filtration.

C. Assessment of the system's performance and accuracy using machine learning algorithms:

Explains the machine learning algorithms used to analyse sensor data and evaluate system performance. The study also looks at the metrics used to determine the system's precision. Additionally, the accuracy assessment results are presented. This study focuses on precision, recall, F1 score, and other relevant metrics. Furthermore, it looks into and acknowledges the strengths and limitations of the machine learning approach employed.

D. Comparison of the achieved results with established air quality standards and guidelines:

To determine air quality, compare the collected data to predefined criteria and norms. This allows us to see if the levels of toxins in the air meet the criteria set by respected organizations like the Environmental Protection Agency (EPA) and the World Health Organization (WHO). This allows us to make more educated decisions about present air quality levels and their possible impact on public health. Assesses the system's performance against established thresholds and standards. The article discusses the impact of the findings on human health and the environment, as well as opportunities for improvement to meet air quality standards.

Fig. 7 and Fig. 8 represent the front and rear view of the proposed system respectively.

V. CONCLUSION AND FUTURE SCOPE

In our proposed model, the real time air quality will be monitored and filtered using MQ-02 and MQ-135 gas sensors, temperature and humidity sensors. The system depends on



Fig. 8. Rear View of Proposed System

the terrain to filter the polluted air. The system uses Arduino and GSM modules to demonstrate air pollution detection and removal capabilities, as well as real-time monitoring and data collection. If any change in the level of the polluted air then it sends a alert message(SMS) through GSM module. The machine learning algorithm's integration allows for the accurate analysis of sensor data and the evaluation of weather accuracy. The results show that the application meets or exceeds air quality analysis.

Future plans include free energy through solar energy and further system optimisation, the exploration of new technologies, the development of data analysis algorithms, and the expansion of monitoring to large or multiple areas for air quality management.

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